

**Base policy
and observable state**



**Six intervention players
repeat, explore, tail, diversity,
novelty, provider balance**



**Canonical composition
and all $2^6 = 64$ coalitions**



**Shared-seed rollouts
utility, relevance, fatigue,
and provider exposure**



**Exact Shapley regions
and interactions**



**Robust planner
improve, repair, or abstain**



**Portfolio decision
and explanation card**



**External ranking audit
shared candidates; frozen tests**